MILITARY SPECIFICATION SHEET
SYNCHRO TORQUE RECEIVER, TYPE 26V-10TR4

This Specification is Mandatory for use by all Departments and Agencies of the Department of Defense.

(This Synchro is for replacement purposes only, not recommended for new design.)

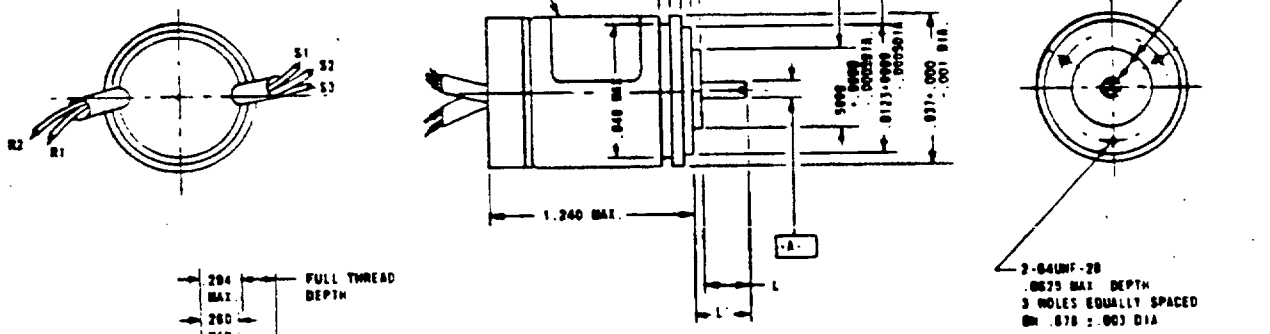
THE COMPLETE REQUIREMENTS FOR PROCURING THE SYNCHRO DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF MIL-S-20708.

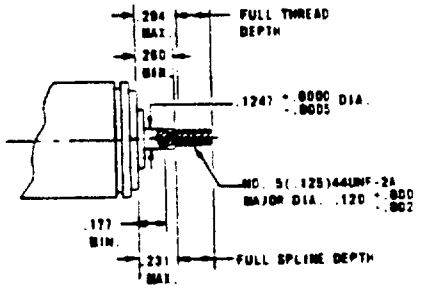
SPLINE DATA	
TEETH	13
PITCH	120
PITCH DIAMETER	.1083 +.0000 -.0020
OUTSIDE DIAMETER	.1241 +.0000 -.0005
PRESSURE ANGLE	20°
ROOT DIAMETER MAX	.089
TOOTH FORM	FULL DEPTH INVOLUTE

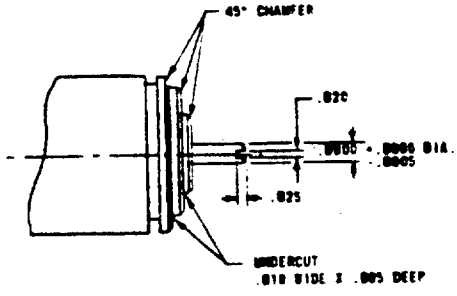
NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE MS-90406-2

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCE ON FRACTIONS ± 1/64
DECIMALS ±.005 ANGLES ±9°

MARK ON HOUSING IS TO MATCH SHAFT MARK WITHIN 10° WHEN ROTOR IS SET AT APPROXIMATELY SYNCHRO ZERO, AND WILL BEAR NO RELATION TO THE CENTER LINES SHOWN ON THIS DRAWING.







NOTES

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING ROTATED.
2. RUNOUT OF FEATURE 'A' (SMOOTH PORTION) OF SHAFT SHALL NOT EXCEED .0005 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING REVERSAL OF THE FOLLOWING LOADS
   END PLAY - 1/2 POUND RADIAL PLAY - 1/2 POUND
4. TERMINAL LEAD WIRES SHALL DEPART THROUGH THE BACK SURFACE OF THE TERMINAL END OF THE SYNCHRO IN ANY ACCEPTABLE MANNER
5. THE FOLLOWING HARDWARE, WHEN REQUIRED, MS-17186-3 DRIVE WASHER (1 REQUIRED) AND MS-17187-1 NUT (1 REQUIRED) SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS. THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SYNCHRO PACKAGE
6. MINOR VARIATIONS OF UNIT CONFIGURATION ARE PERMITTED FOR UNDIMENSIONED DESIGN DETAIL.
7. HOUSING SHALL BE OF ALUMINUM.

FSC 5995

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TABLE I

Requirement	Value	Unit	Tolerance
Frequency	400	Hz	±1%
Primary Voltage	26	volts	±1%
Primary Current	142	milliamps	maximum
Primary Power	.62	watts	nominal
Impedance: Zss	11	----	nominal
Impedance Angle: Zss	15	----	nominal
Transformation Ratio	.456	----	±2%
Phase Shift	7.5	degrees	maximum
Receiver Error	60	minutes	maximum
Synchronizing Time 30° ±2°	2	seconds	maximum
Synchronizing Time 77° ±2°	4	seconds	maximum
Torque Gradient	2200	mg-mm	minimum
Radial Play	.001	inches	maximum
End Play	.002 - .005	inches	min.-max.
Temperature Rise	20.0	degrees (C)	maximum
Variation of Voltage (+10%) and Freq. (-5%)	1.4	watts	maximum

Part number: M20708/500 plus applicable dash number from Table II.

TABLE II

DASH NO.	SHAFT DIAMETER	SHAFT STYLE	TERMINAL END	L' DIMENSION	L REFERENCE FREE SHAFT LENGTH
001	.0900	Plain & Slotted	18" Wire Leads	.321 ± .015	.283 ± .015
002	.1247	Splined	18" Wire Leads	.500 ± .015	.462 ± .015

Custodians:
Army-MU
Navy-AS
Air Force-80

Review Activities:
Army-EL,MI,MU,ME,AT,AV
Navy-SH,AS
Air Force-11

User Activities:
Army
Navy-SH,MC,GG
Air Force-11,80

Preparing Activities:
Navy-AS

Project No. 5990-0297

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